Thread - 2025-02-18

https://x.com/RiikonenMarko/status/1891891551682347050

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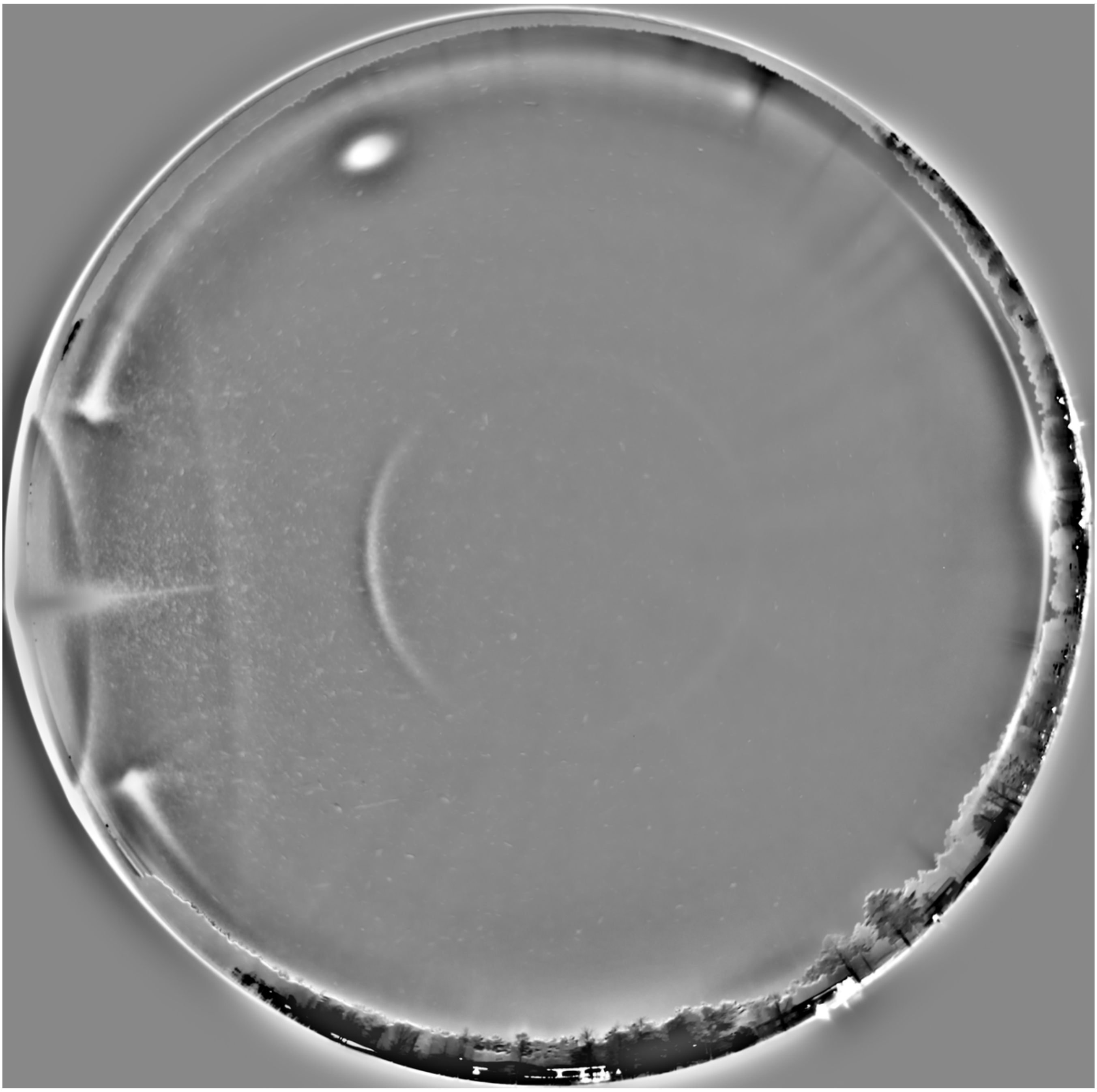
The night of 15/16 February 2025 in Rovaniemi. I had waited at the apartment for the snowfall to stop. It was pretty much calm, so action was expected. When I got going around 2 am, Ounasvaara webcam already had pillars on the slope lights.

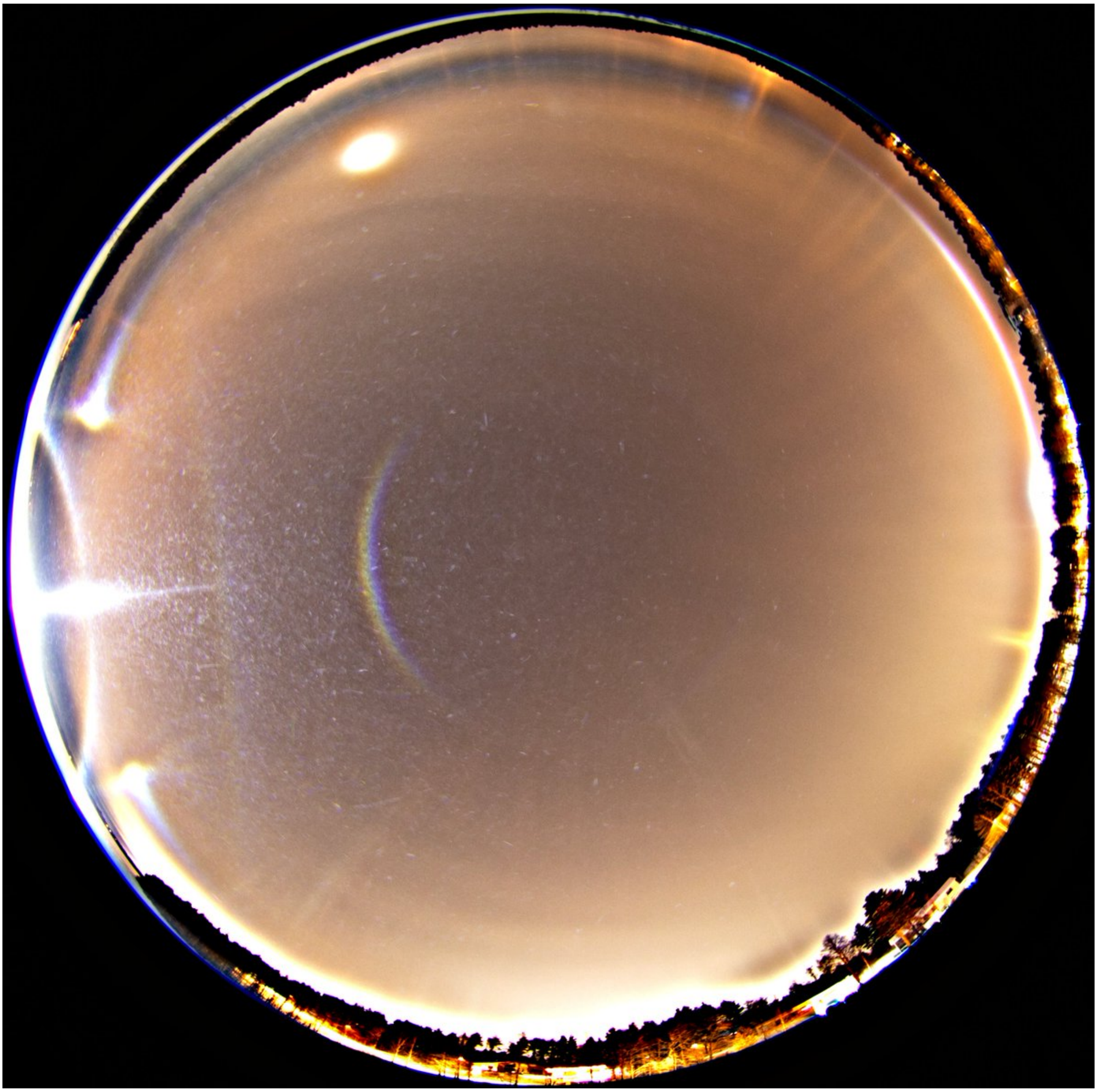
I ended up at the eastern tip of the lake Salmijärvi. There is an open landfill and also now a single earthen mound to allow for extra negative lamp. As I arrived, Moon was visible for a while giving a display. I got it photographed by putting one camera on the car roof while I set up the spotlight.

In the beam at first there was nice flip Liljequist parhelia, so I took all-sky view. Photos also captured flip Dufay (Dufay is the right name for any Kern named arcs, details will follow later in the nomenclature article) and flip 120° parhelion. The last image here is a max stack and it does not show flip Dufay, which is in line with me not being able to see it. However, the flip 120° parhelion shows up in max stack, meaning I should have seen it had I looked.

I am using term 'flip' instead of 'counter' that I have employed earlier in place of 'sub' and 'super'. Just getting a feel how they rhyme. There is already a precedent for this term: Jia Hao used 'flipped' four times in his The Halo Vault blog post "Hidden Treasures in the Jutland Display".

The car thermometer was at -13 C, which is right at the divide the warmer side of which big plate is good prospect and colder side crap. The official figure at the railway station at the time of these photos around 0217 was around -12 C. This was fog / stratus turning into ice crystals.





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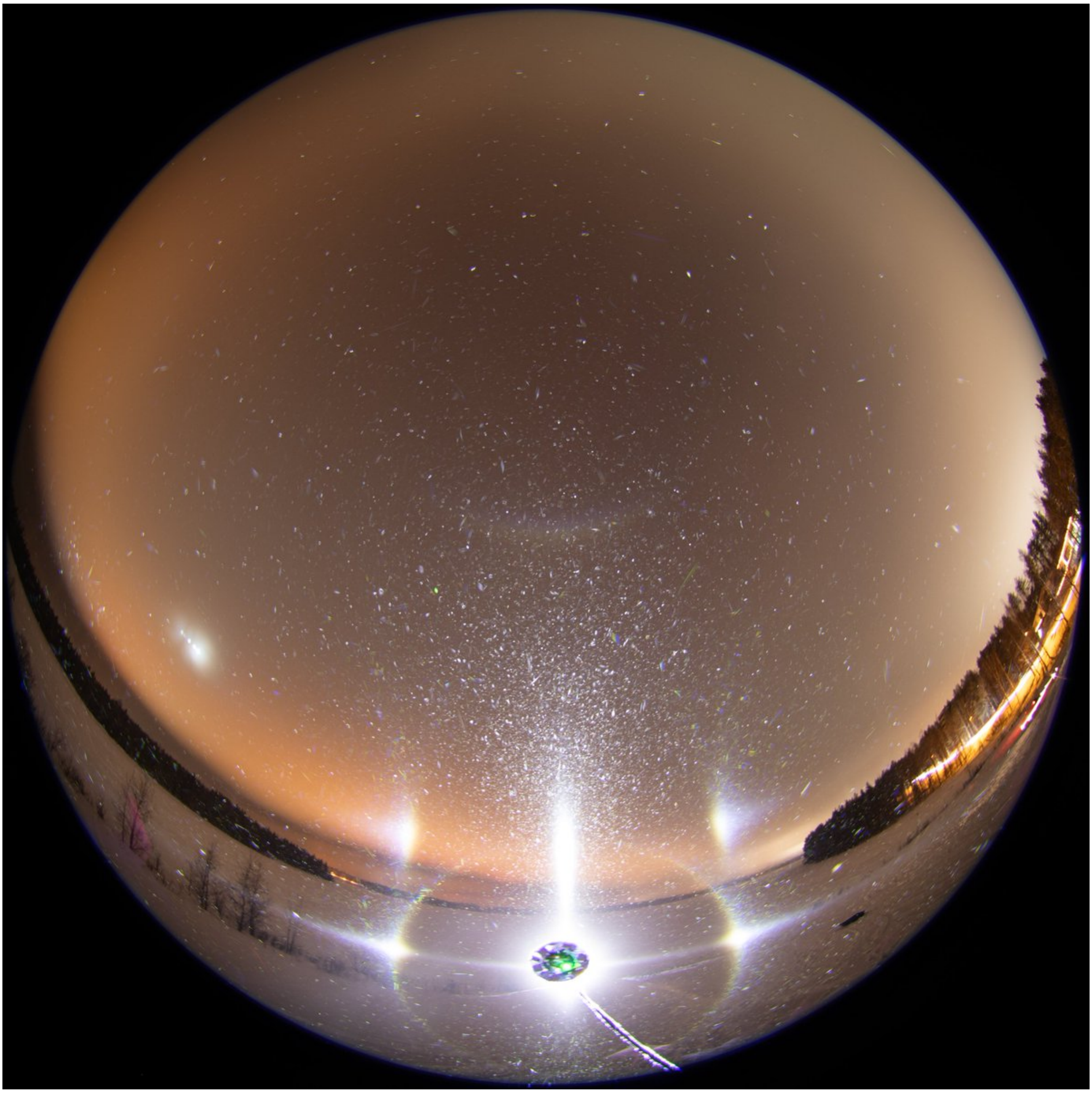
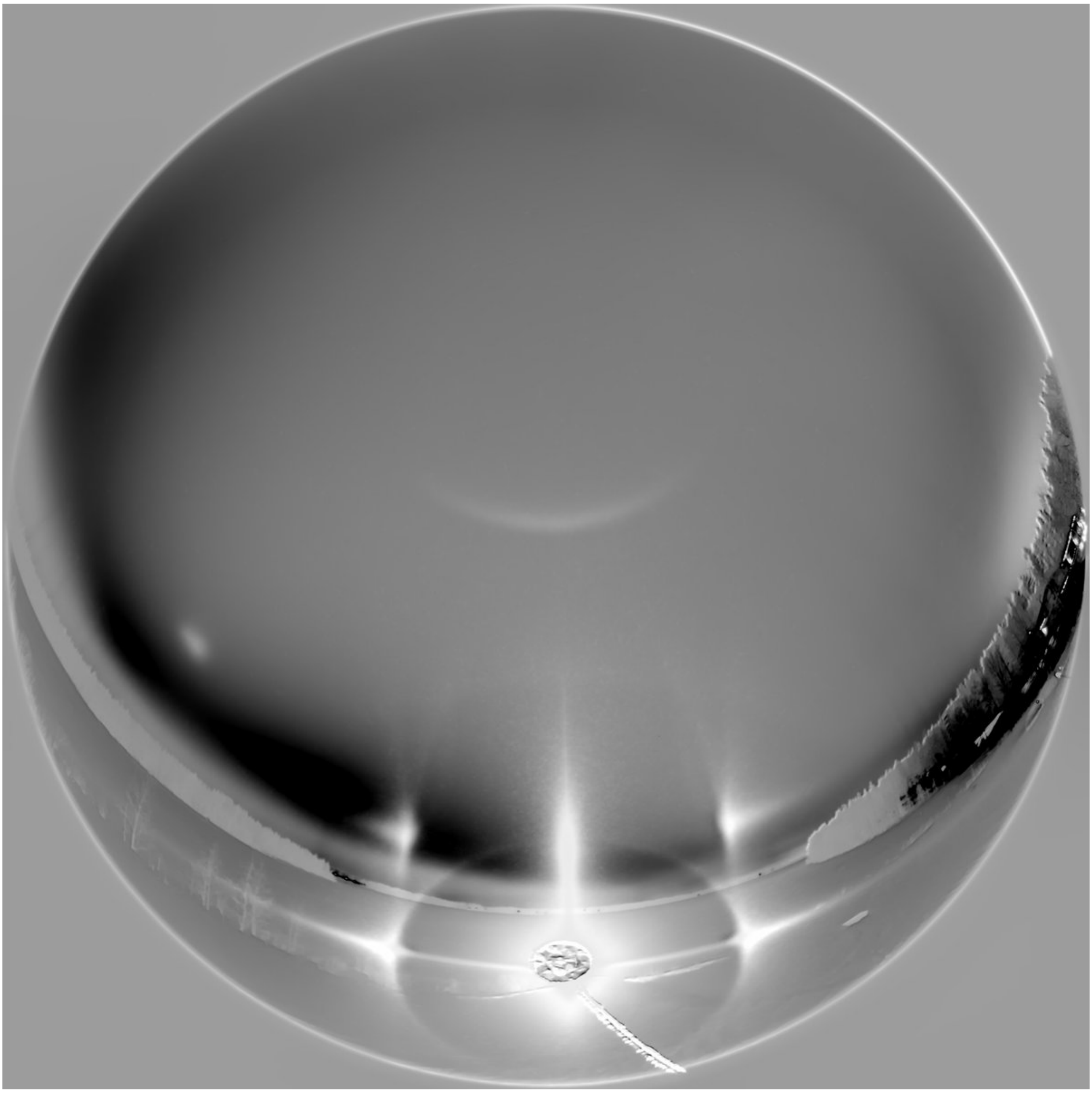
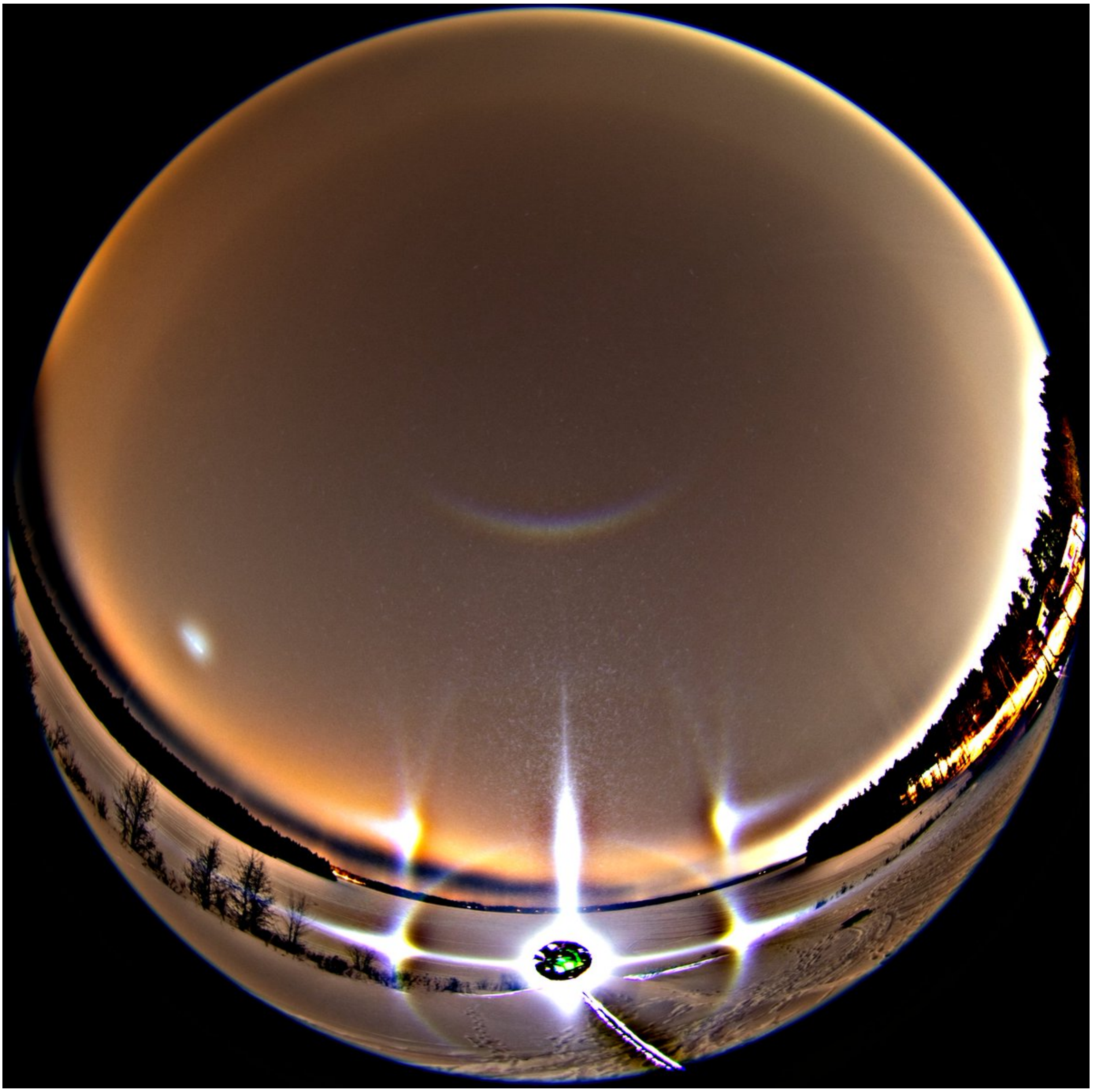
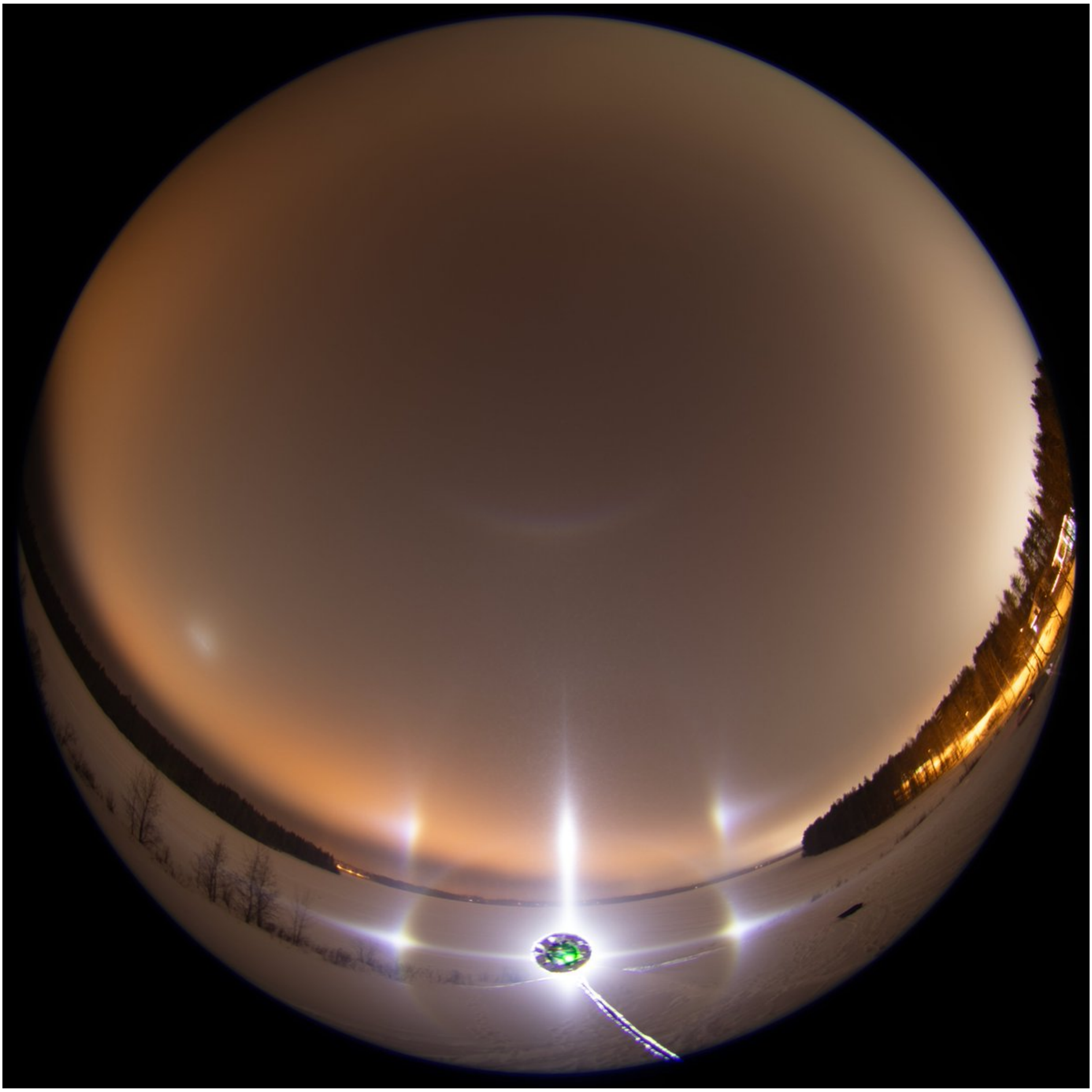
15/16 February 2025 Rovaniemi. The display then practically receded to the lamp side only and I adjusted the camera accordingly. This is quite long stack, 59 x 15s. The two arcs associated with flip parhelia are somewhat better now and weakly visible is also the spatial instance of the flip parhelia. The lamp, Olight Javelot Turbo, has a cone of flood surrounding the bright central spot and this is the cause of the 3D instance.









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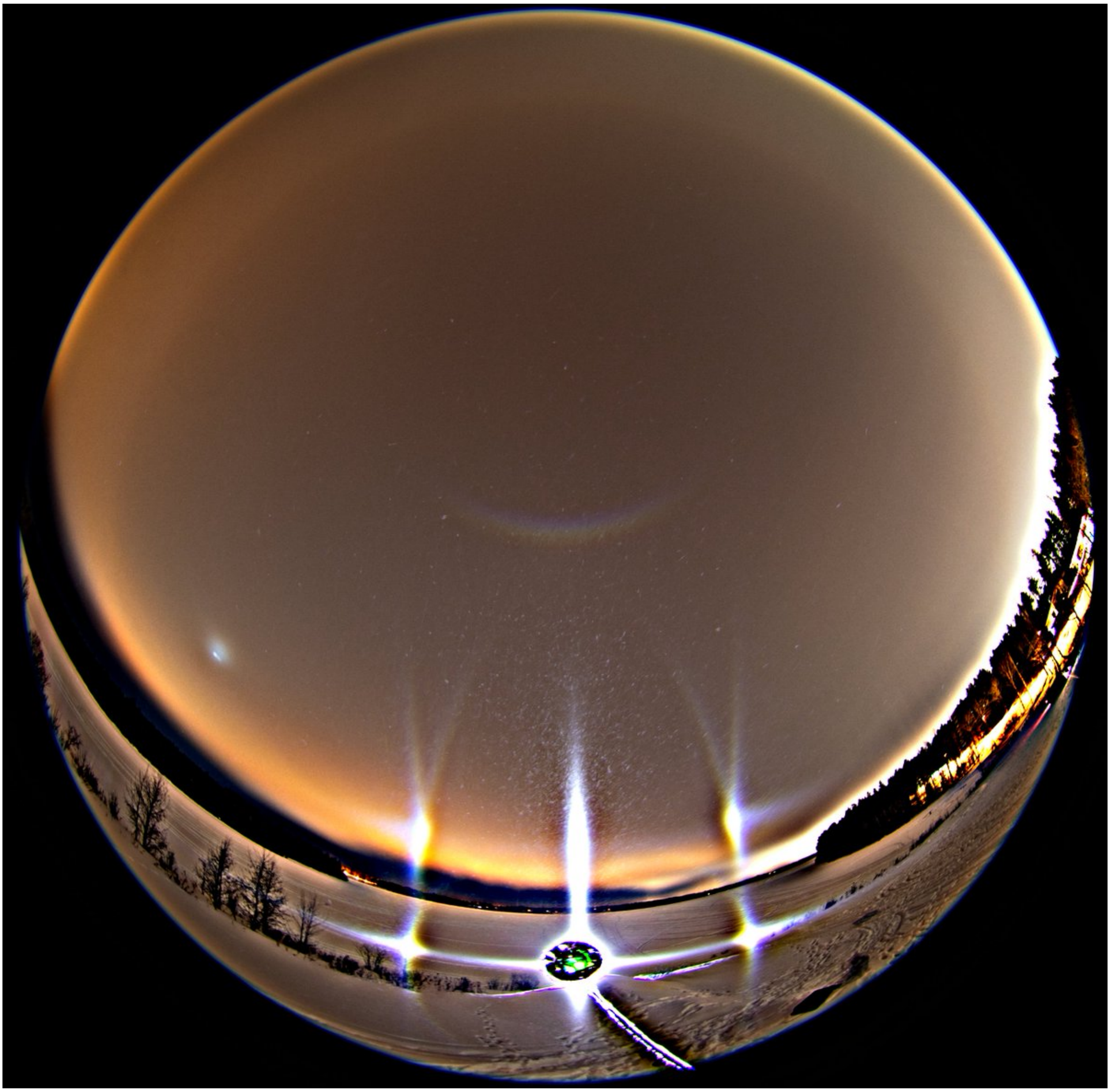
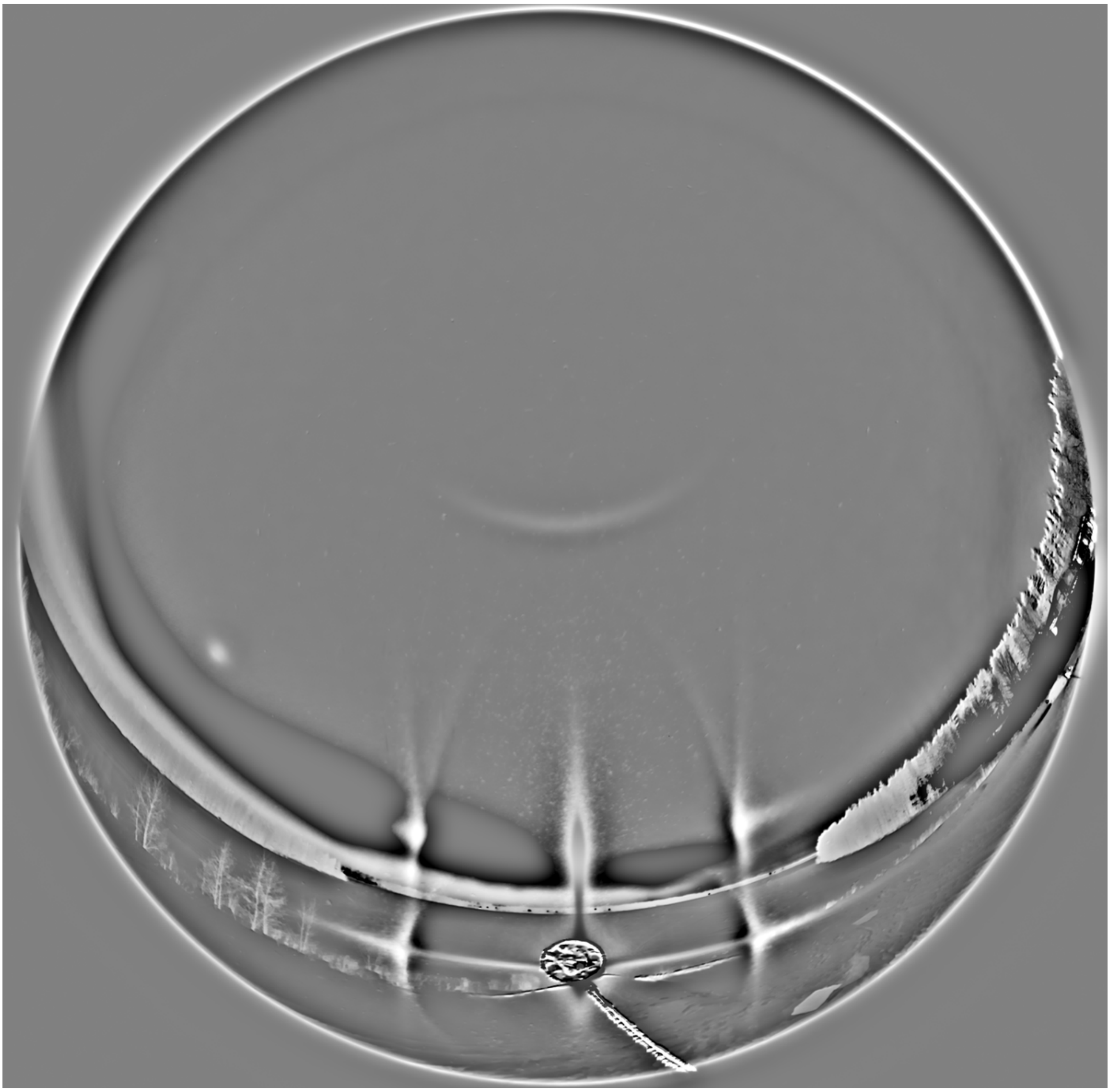
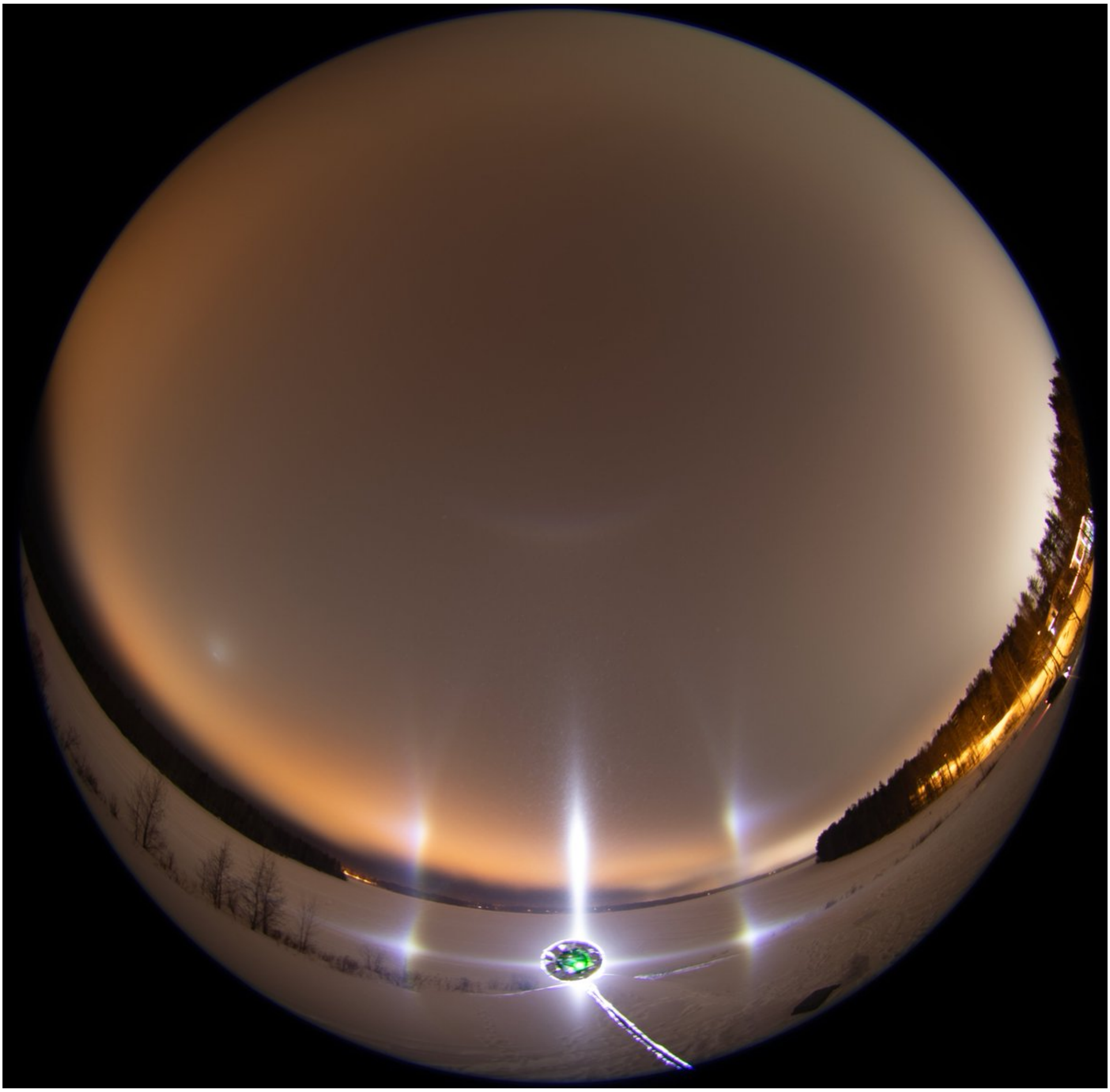
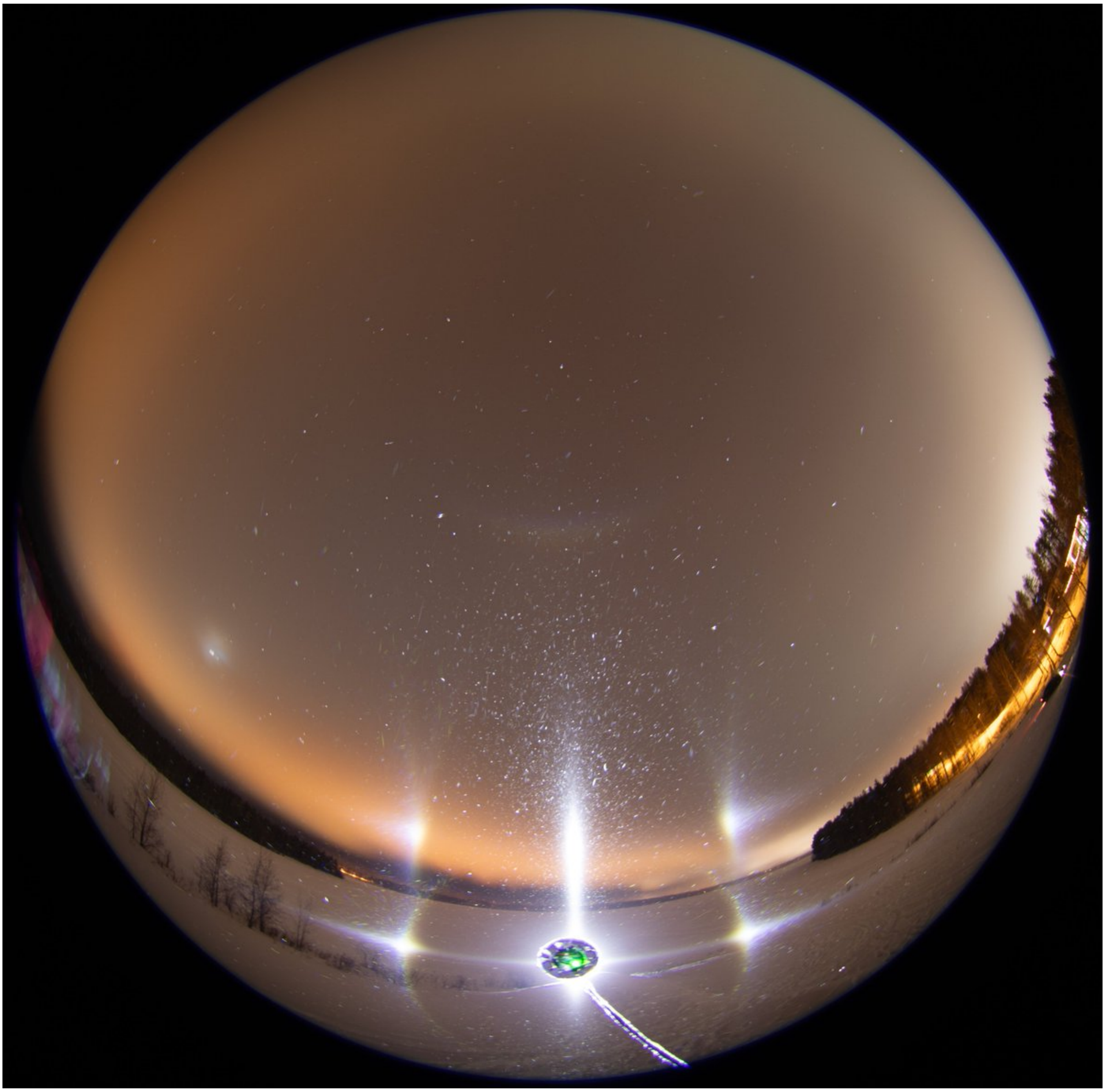
15/16 February 2025 Rovaniemi. In the previous stack the individual shots did not consistently have the flip parhelia associated arcs, but then a clear change happened and they became a prominent fixture. This stage lasted 21 frames and after which the display started fizzling out. No more good action was on the offing for the rest of the night.

The arcs associated with flip parhelia can't be explained with Lowitz theory. It does not give the right curvature. Simulation would also produce 46° contact arcs and very strong Lowitz arcs, which are not visible.









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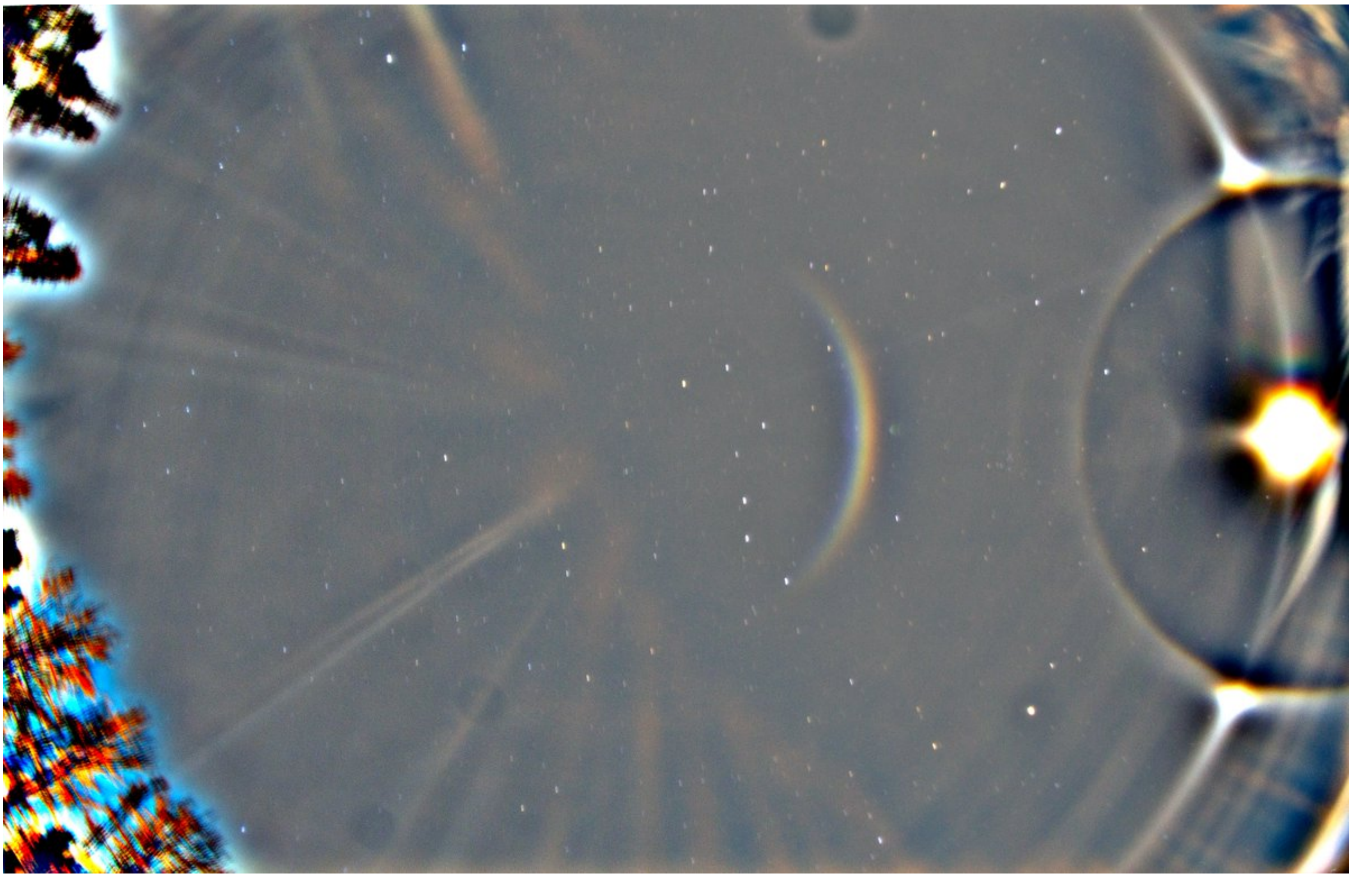
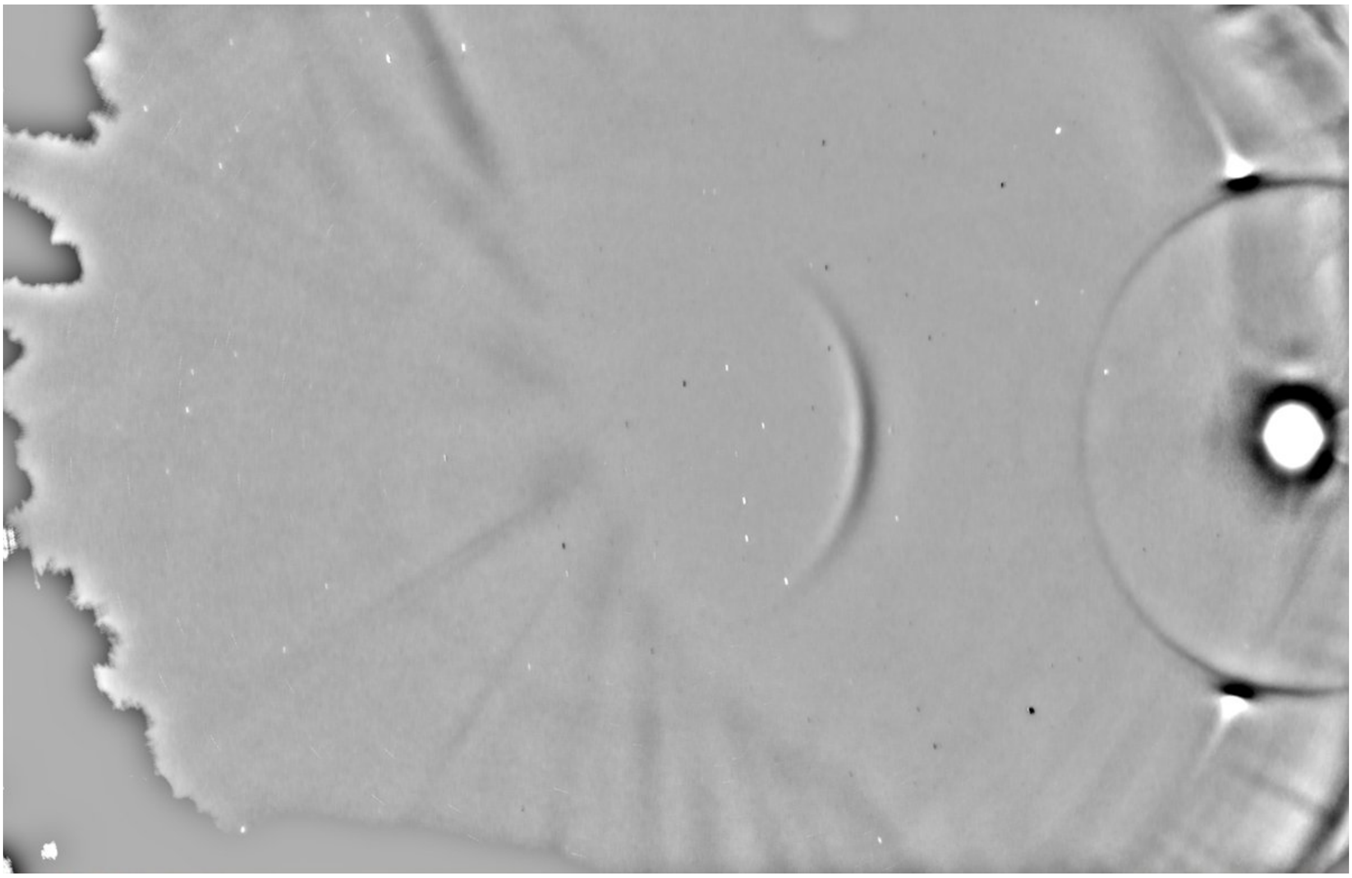
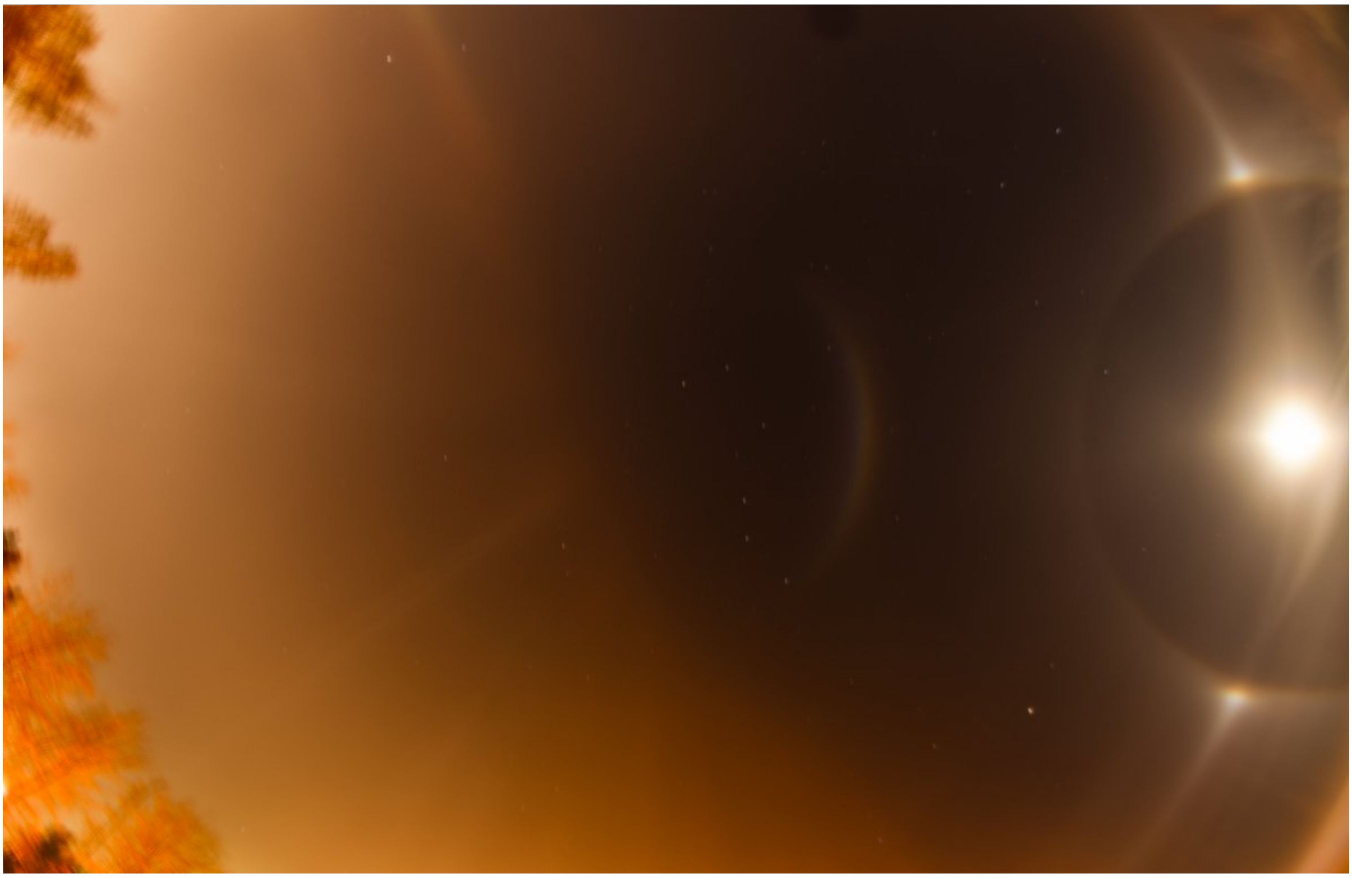
15/16 February 2025 in Rovaniemi. I got 10 x 30s of the lunar halos with the car roof camera before the moon was gone behind thickening cloud. I manually rotated the images to align the display for the stack. The Parry arc is of interest because it occurs without any indication of column orientation halos whatsoever. I don't know much such displays, if at all.







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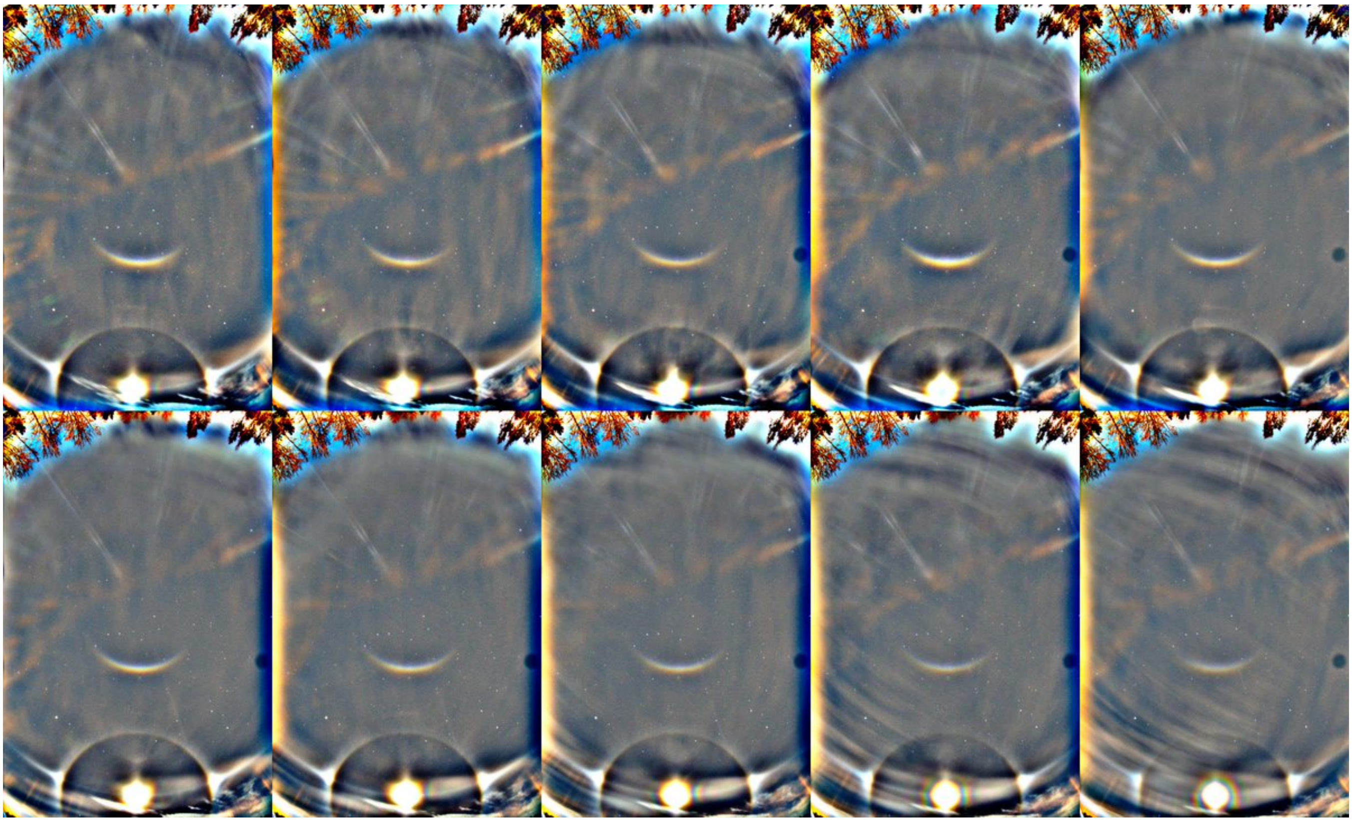
15/16 February 2025 in Rovaniemi. Just to make sure the Parry is not lens artefact, I looked at all the individual photos. There is variation in the arc that would be hard to explain as an artefact.



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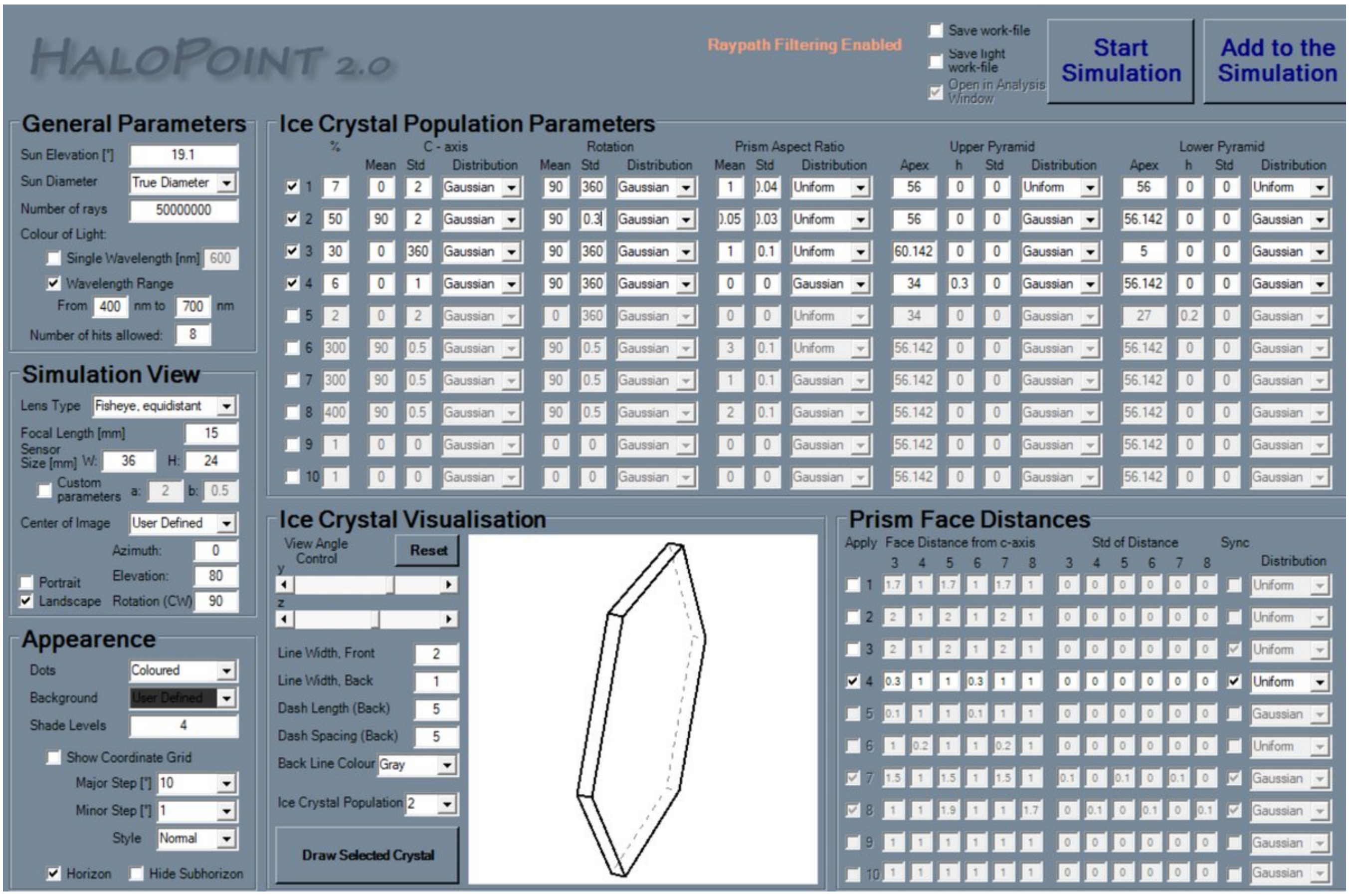
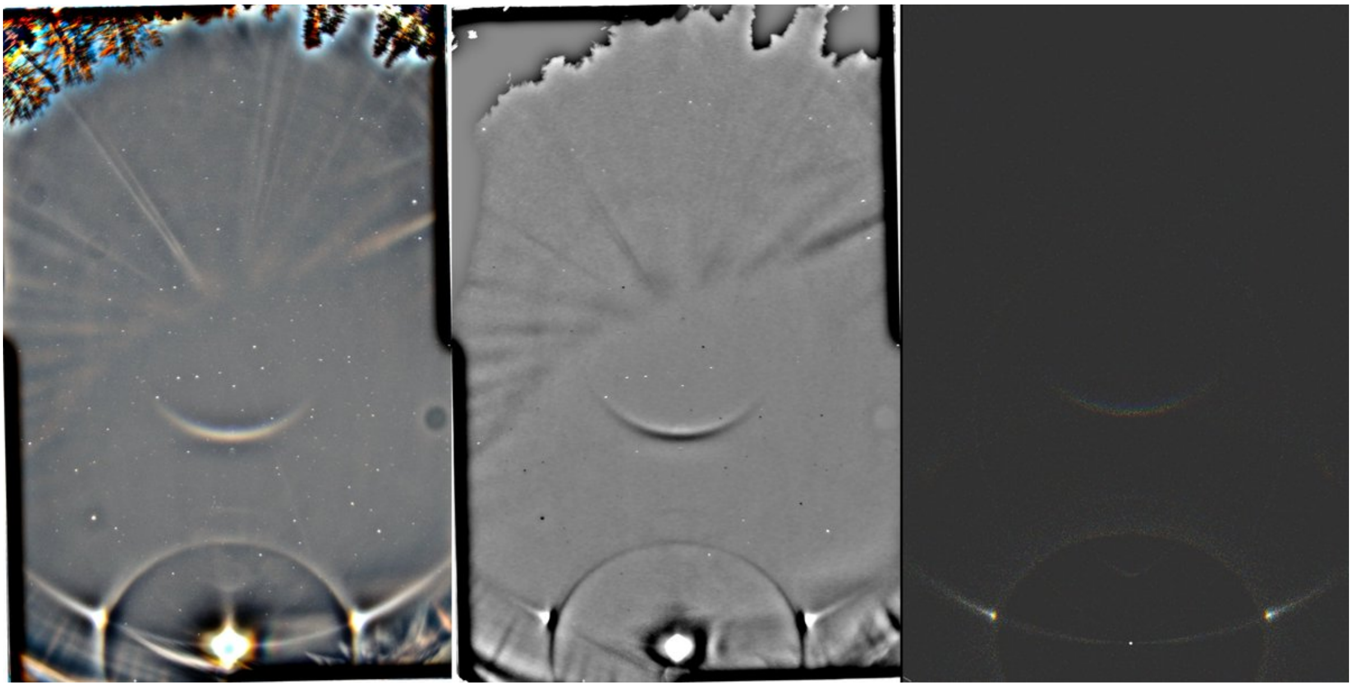
15/16 February 2025 in Rovaniemi. Also simulated Parry position agrees, lunar elevation 19.1 degs. I included also the Moilanen arc. The two versions of the display here have 8 frames stacked, two last frames omitted because they did not have the Parry. I used very thin plates for the Parry arc and two degrees tilting about the Lowitz axis, so maybe that is rather Lowitz arc than Parry.





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Now I think this case is too weak to say there is no tangent arc. 22° halo is actually a bit more intense in the stack at the top compared to the oblique locations. This is supported also by the single photos. I think what we have here is similar to the display on 19 January 2021 in Chechia: https://t.co/smNmN3rRM7 only just much weaker and less uniform.

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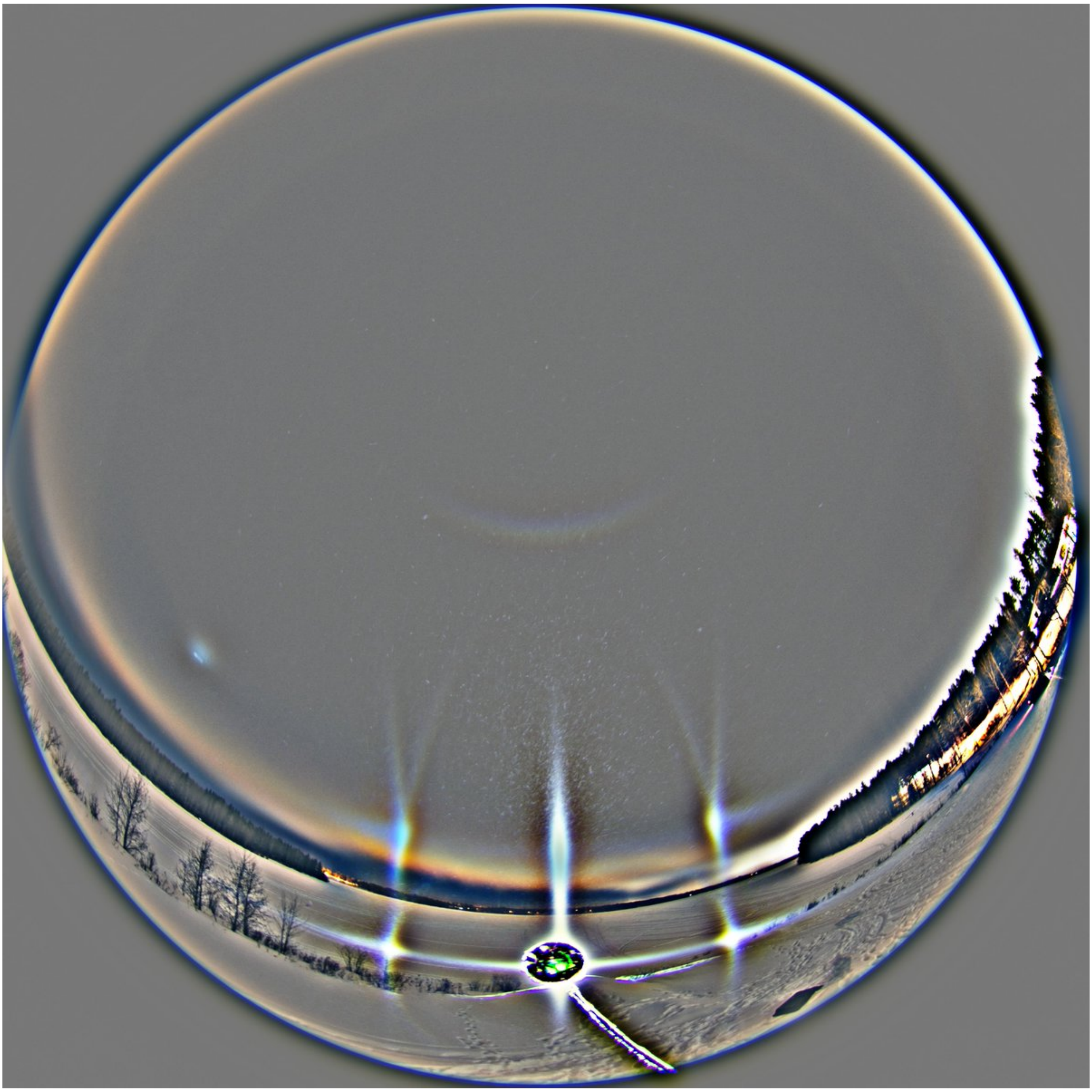
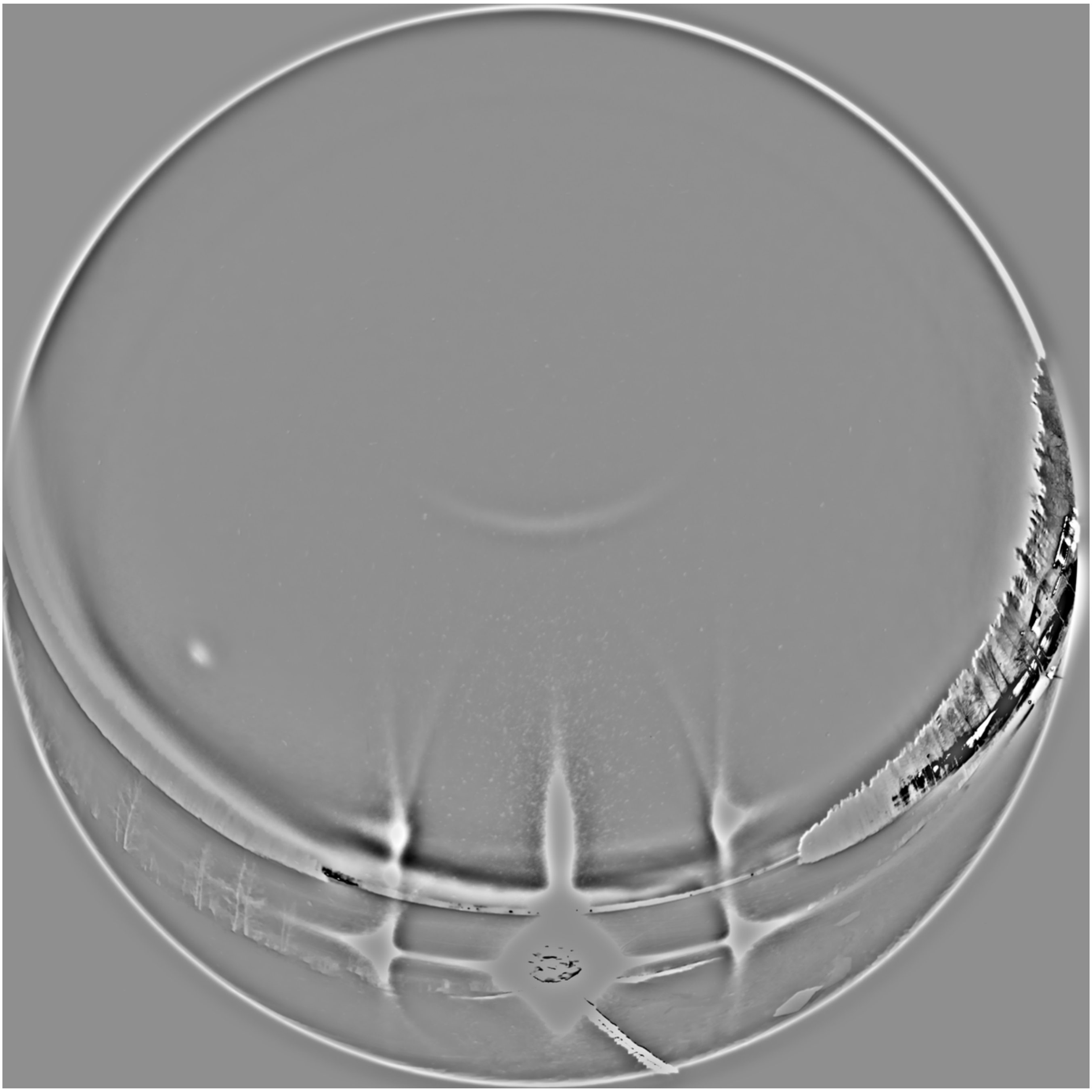
One more thought on this display. I had overlooked the flip parhelic circle segment inside the flip parhelia. The yellow glow of city lights disturbs, but the blue segment is clear enough. Normally this particular part is missing but Lowitz oriented crystals and plate oriented triangular crystals are capable of making it. The former hypothesis has the stronger footing as in all cases the segment is associated with the long flip parhelic arcs. The first of the two greyscale images is the usual br, the second one is bg. Both enhanced with bgr, background removal, as Lefa calls it.





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A video of the 15/16 February 2025 Rovaniemi spotlight display. The unfocused beginning shows the flip parhelic circle and flip Liljequist. Visible in most spotlight displays, but usually weakly. It was brighter than in the video, my low quality phone manages, passably, only the intense stuff on the lamp side.

It is hard work to take these videos right next to the camera as you have to hold your breath to not spoil the photos. Even without videos, you always have to fiddle with the camera and sweaty breath holding is very much part and parcel of the nightly halo chase. There could be some benefits reaped for a haloman to acquaint himself with the techniques of freediving.

[video: 1893696059915845974-xTkSuSVOJs5IK4A4.mp4]

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Another video of the 15/16 February 2025 Rovaniemi display. At this point the stuff on the opposite side had weakened and I had the camera tilted from pointing straight up halfway towards the lamp. As uploading compression on this site tends to wreck havoc on these kind of videos, I have put the originals on Google Drive:
https://t.co/LKCp8mj74I
https://t.co/kg4B1G7VIS

[video: 1893696064147894520-VtIy5qAgk0Qvk3I7.mp4]